

An aerial night photograph of the TU/e University campus in Eindhoven. The image shows several modern buildings with illuminated windows, a prominent red tower, and a large parking lot. A semi-transparent red banner is overlaid across the middle of the image, containing the lecture title.

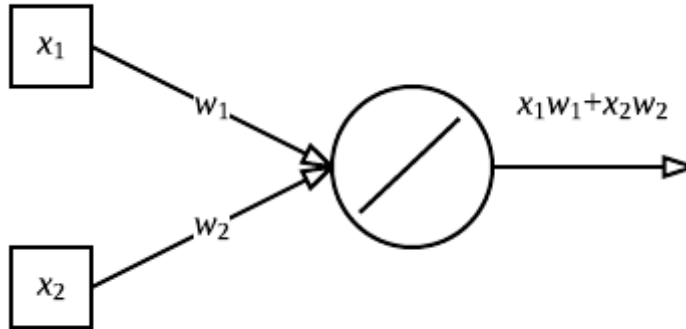
Lecture 8 - Convolutional Neural Networks

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Department of Biomedical Engineering, Medical Image Analysis group

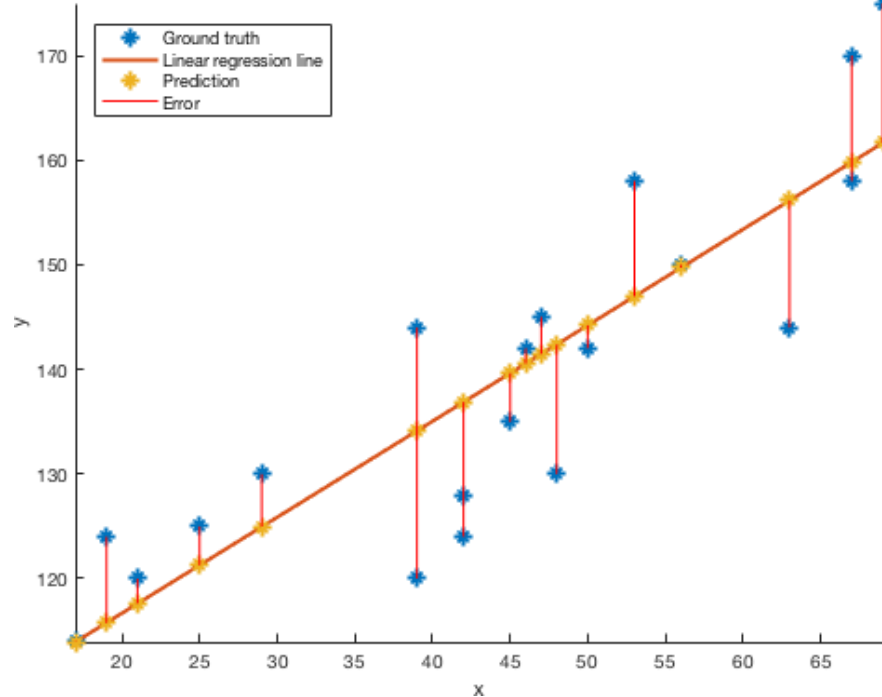
Recap – Machine Learning

Previously – Linear regression

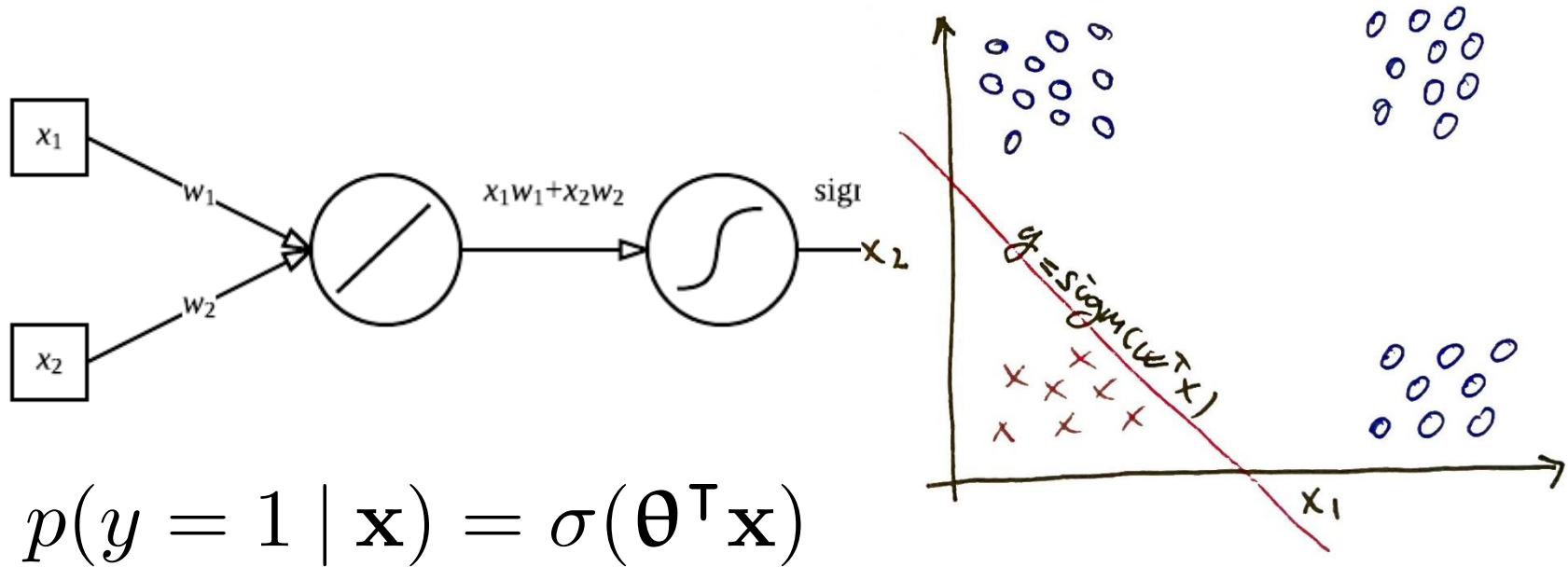


$$f(w_1, w_2, b, x_1, x_2) = w_1x_1 + w_2x_2 + b$$

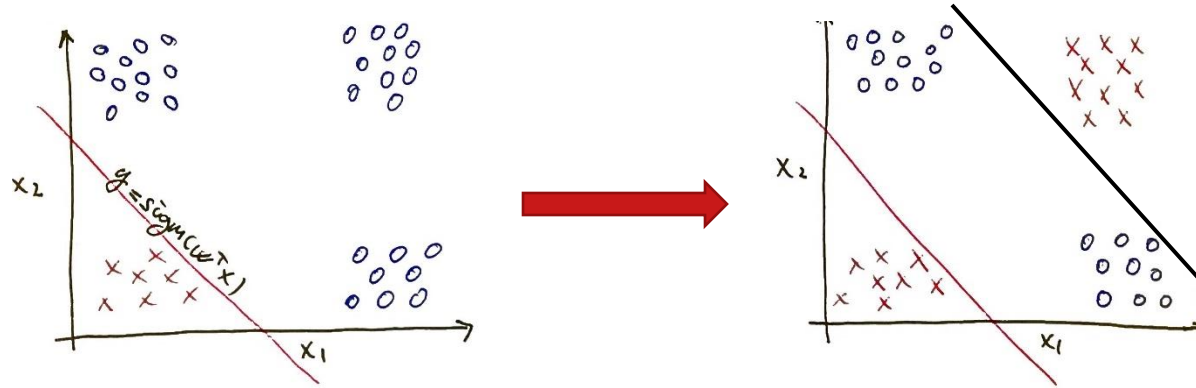
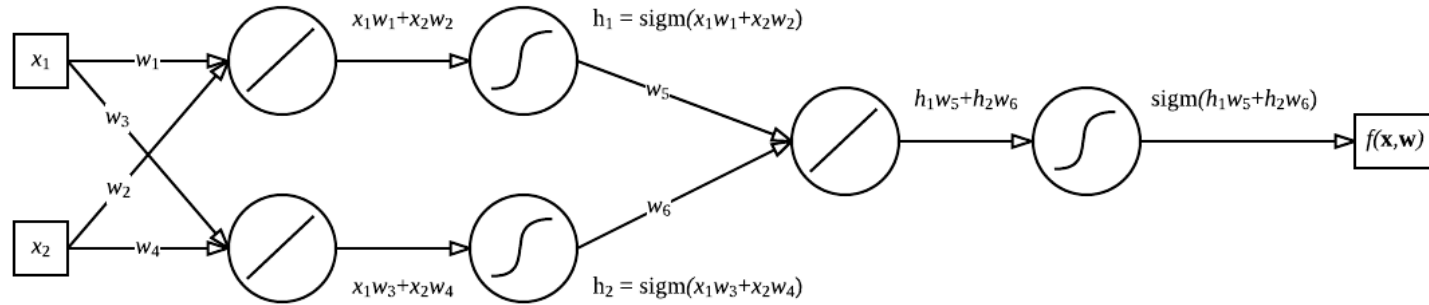
$$\hat{y} = \theta^T \mathbf{x}$$



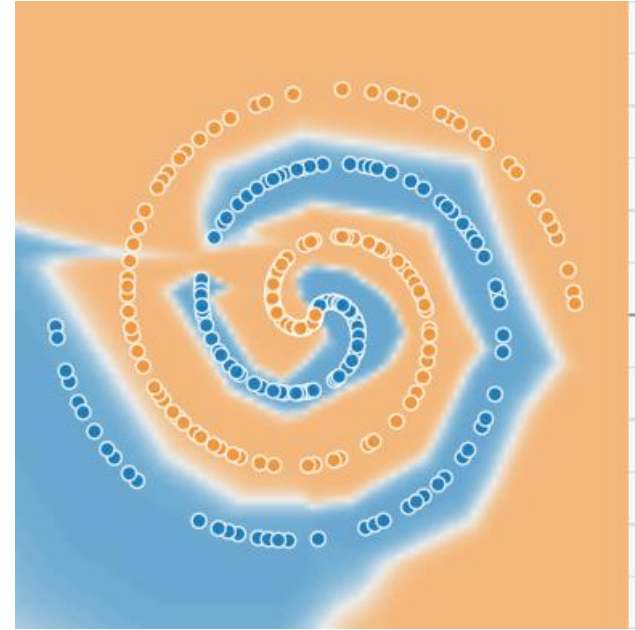
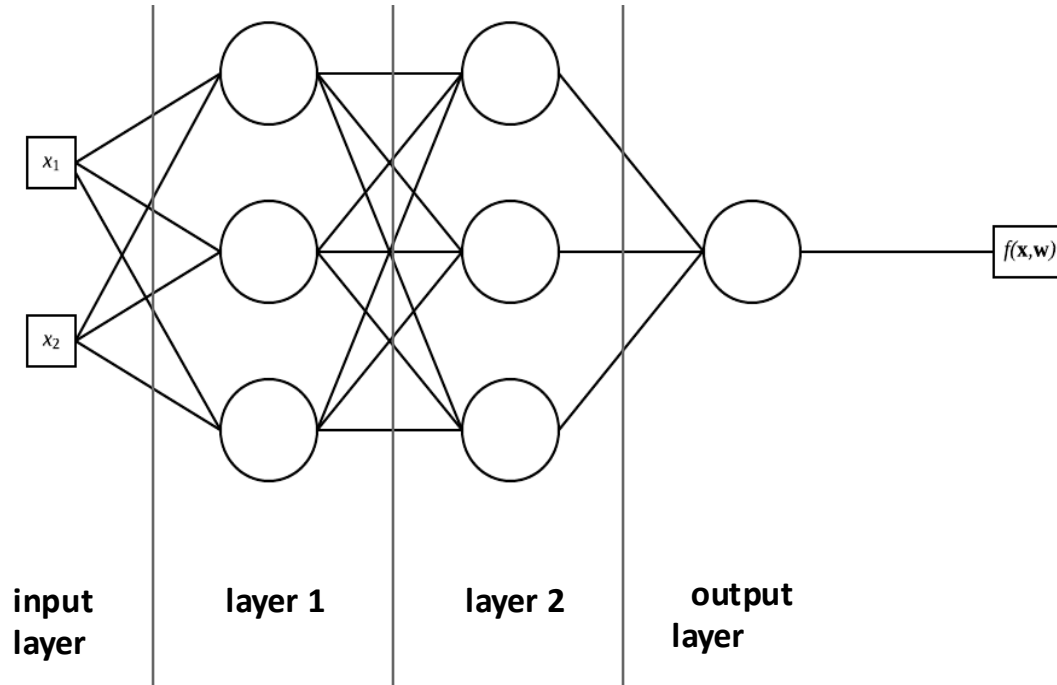
Previously – Logistic regression



Previously – Neural networks

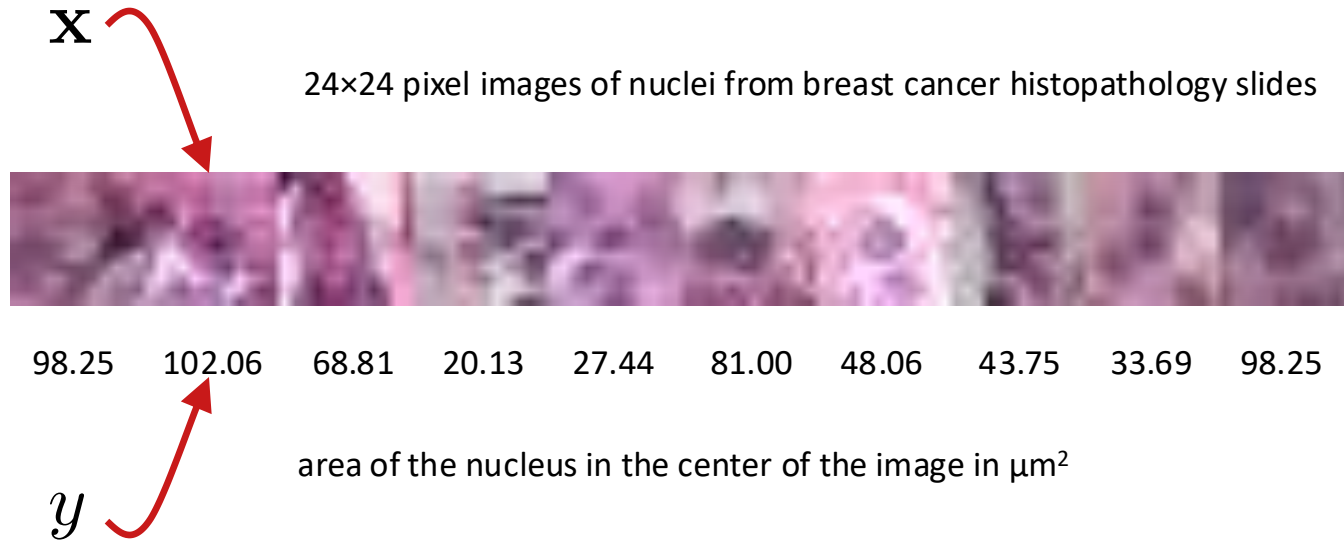


Previously – Deep learning

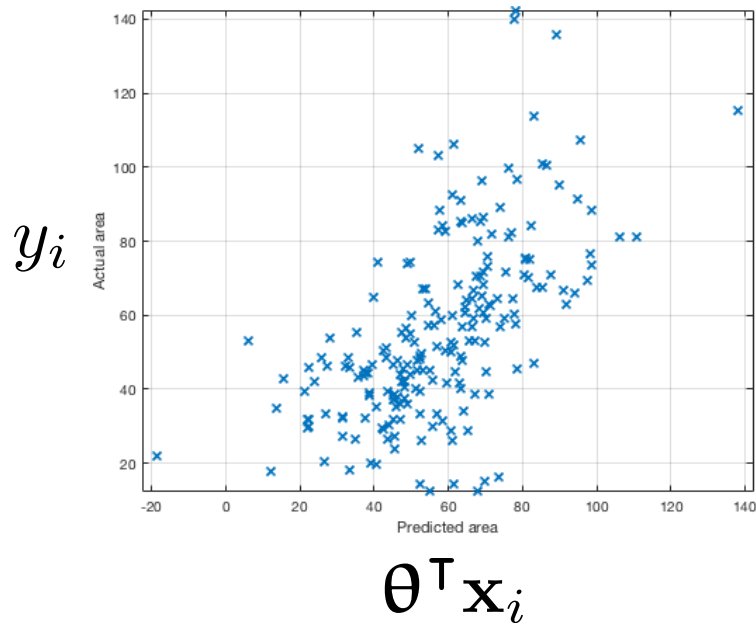


Applied to medical images

Linear regression for medical image analysis- measuring nuclei size:

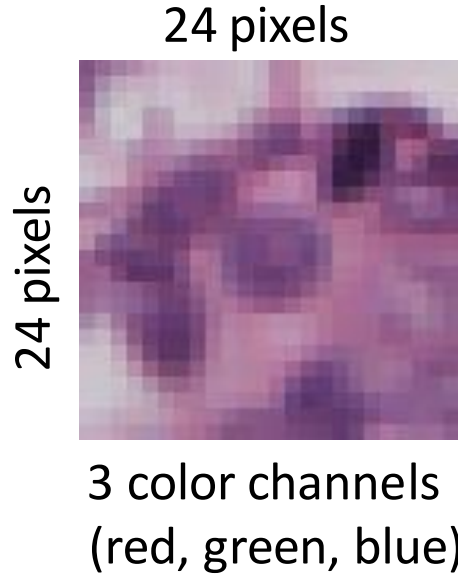


Linear regression for medical image analysis- measuring nuclei size:



NOTE: This is not a plot of
Feature $\mathbf{x}$ vs. target y but
predicted vs. actual y .

Images as input to neural networks



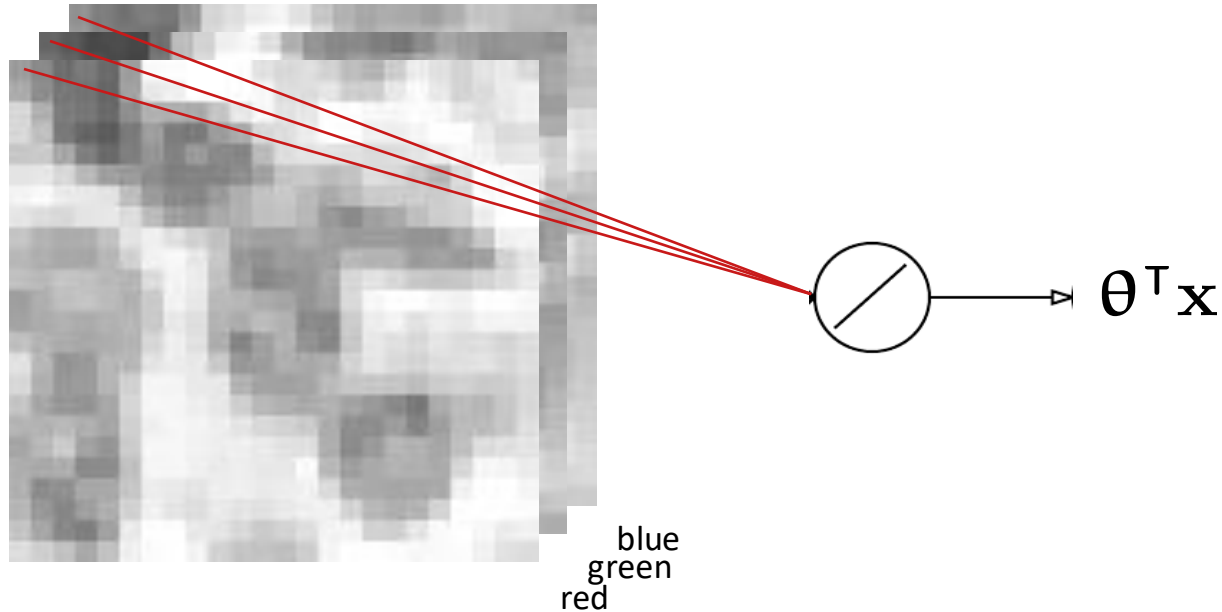
= 1728 features ($\mathbf{x}_i$ are 1728 dimensional vectors)

If we train linear regression with these inputs (such as in the first practical), we will have 1728 weights w and a bias b .

-> 1729 parameters to be learned

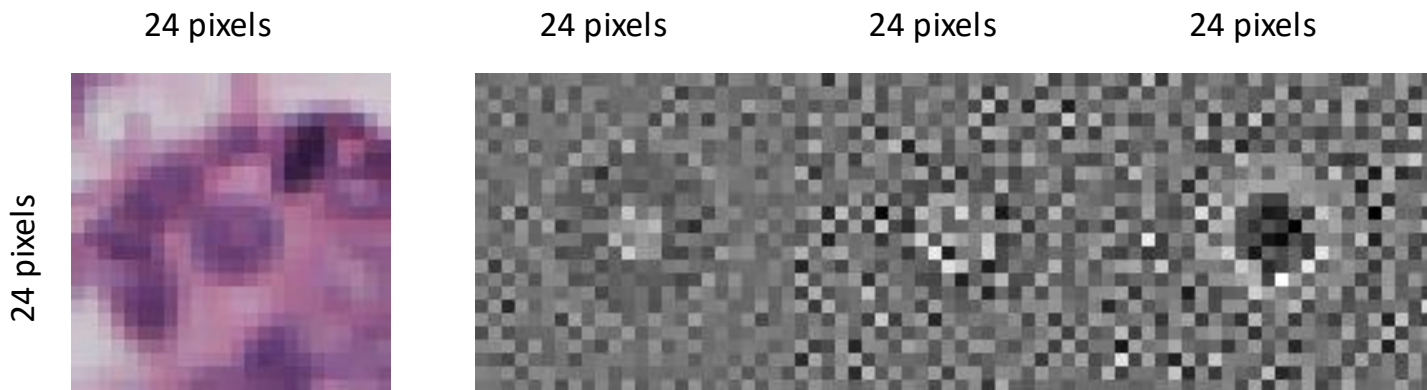
Every pixel is an input

Every pixel from every color channel is multiplied by a weight



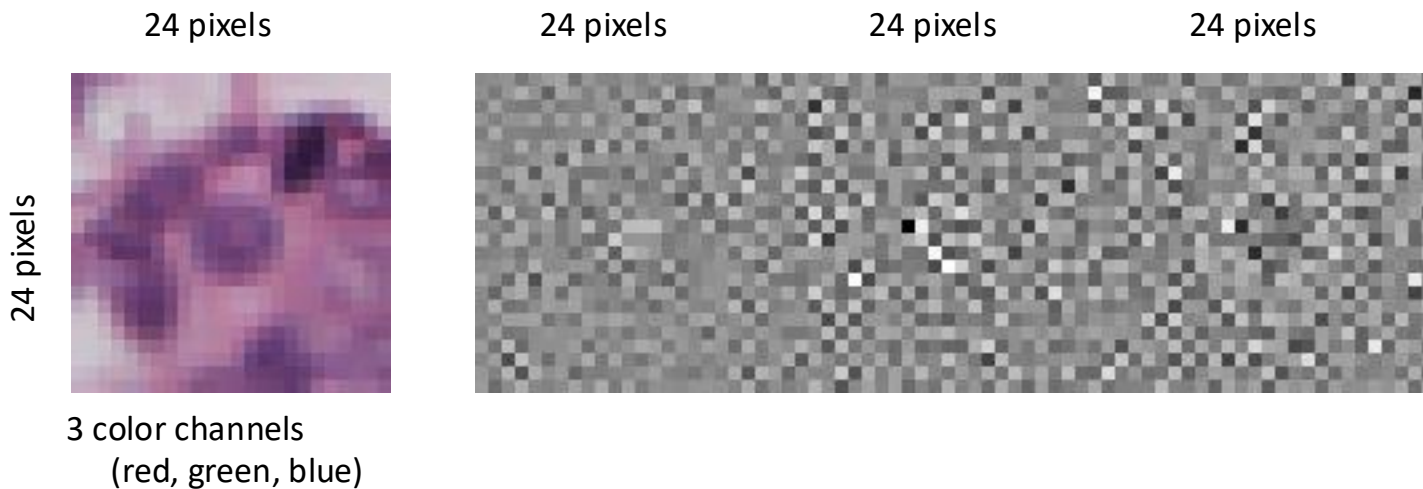
Post-training: visualizing what model has learned

Reshape vector of parameters into 24x24x3 image



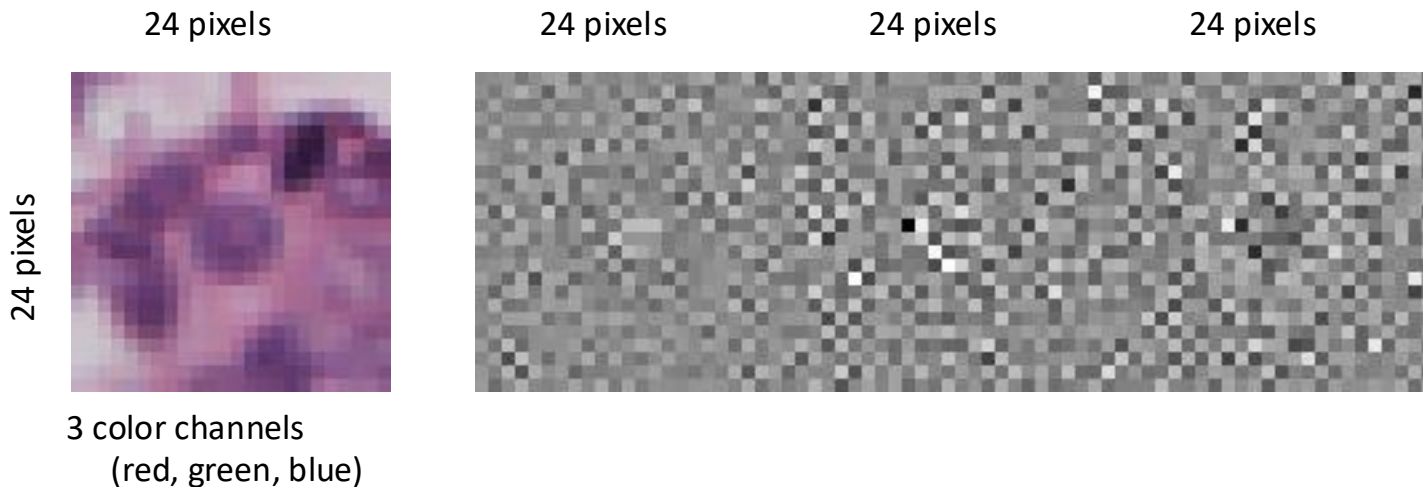
Post-training: visualizing what model has learned

Only 25% of training samples.. Looks noisy!



Post-training: visualizing what model has learned

You can think of it as “there is not enough training data to reliably estimate all model weights”.



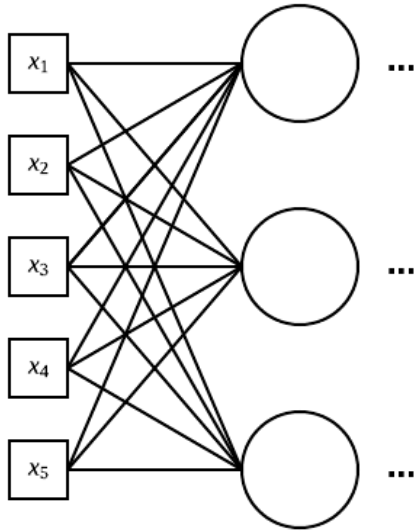
Today: Convolutional neural networks (CNN)

- Neural networks → Convolutional neural networks

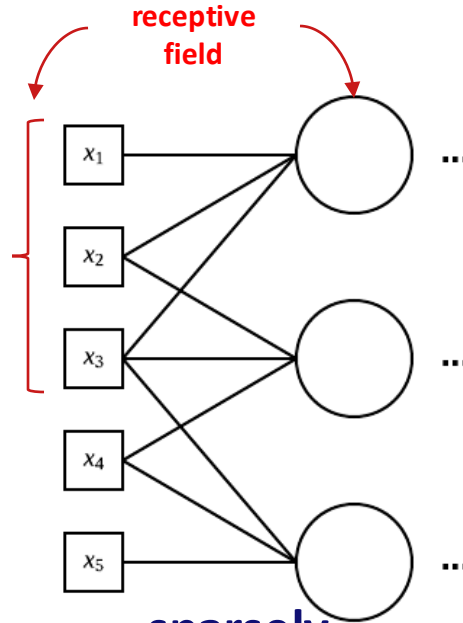
Building blocks for deep learning models for image analysis:

- Convolutional layer
- Max-pooling layer
- Not needed for the project, but will be on the exam

CNNs

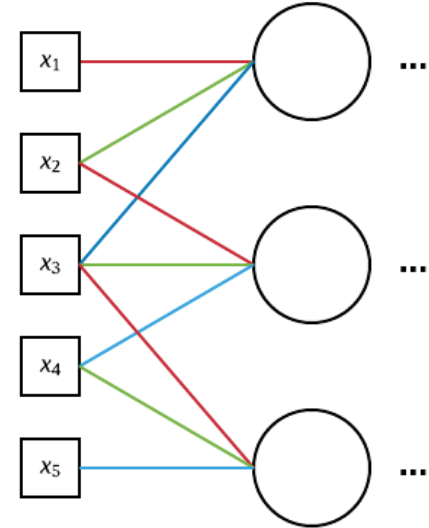


“regular” NN
15 weights



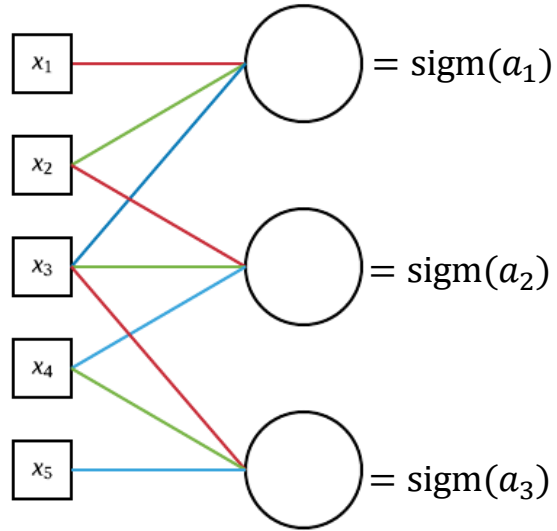
**sparsely
connected NN**
9 weights

How many weights?



shared weights
3 weights

CNNs



shared weights
3 weights

$$a_1 = x_1 w_1 + x_2 w_2 + x_3 w_3$$

$$a_2 = x_2 w_1 + x_3 w_2 + x_4 w_3$$

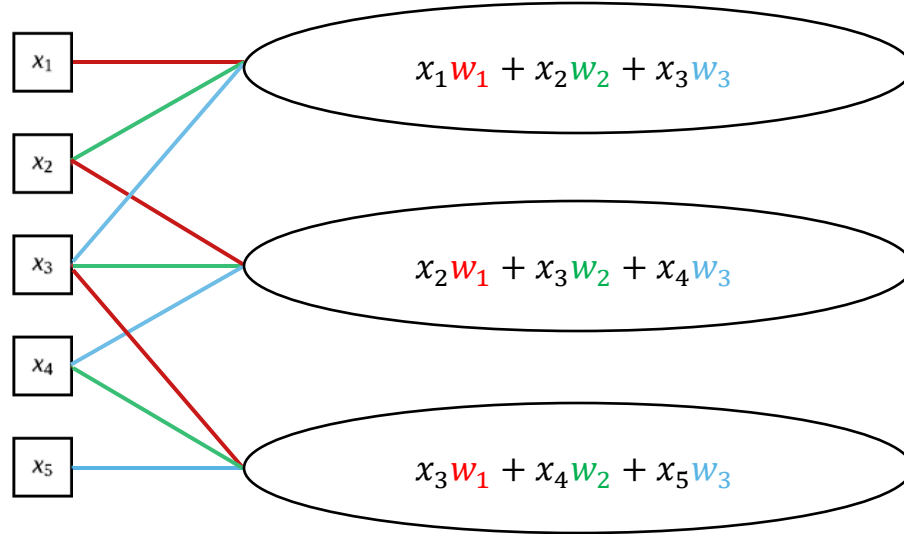
$$a_3 = x_3 w_1 + x_4 w_2 + x_5 w_3$$

$$\begin{bmatrix} x_1 & x_2 & x_3 & x_4 & x_5 \end{bmatrix} * \begin{bmatrix} w_1 & w_2 & w_3 \end{bmatrix} = \begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix}$$

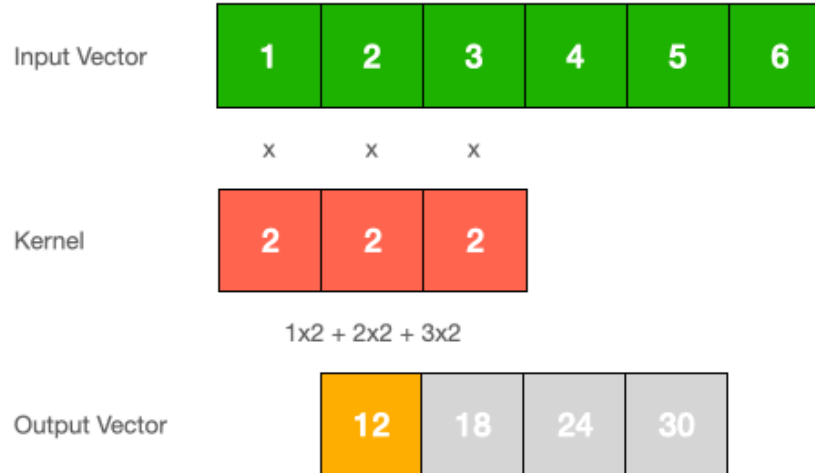
convolution, thus convolutional NN

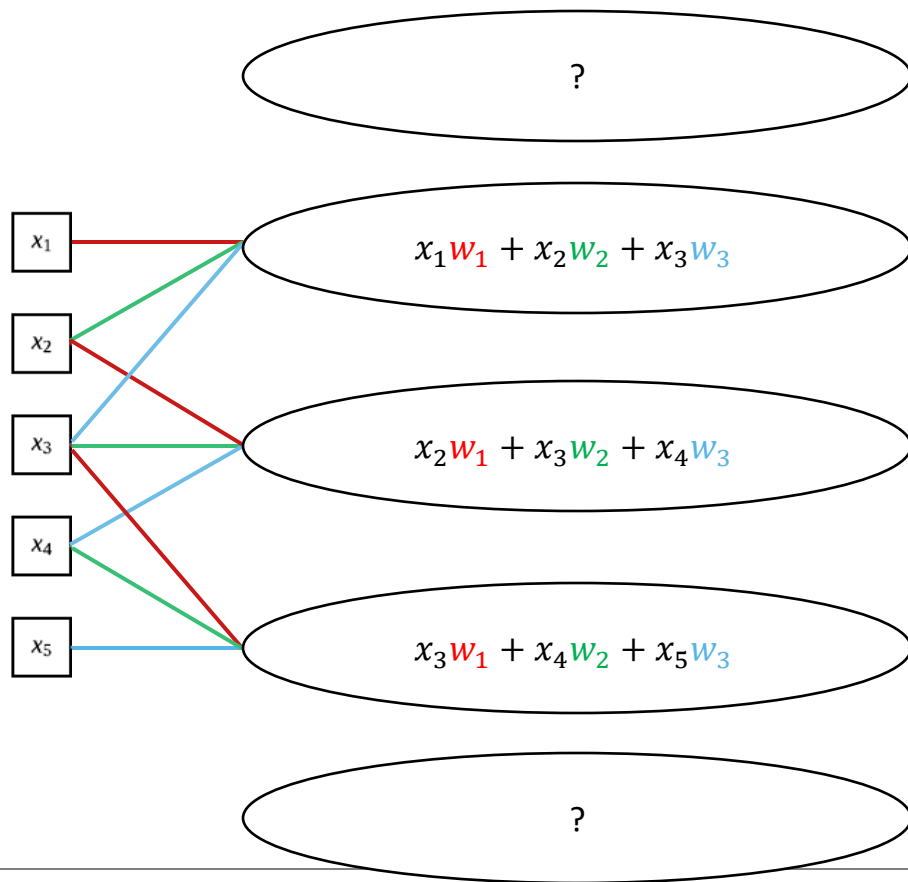
CNNs

1-D Convolution



1-D Convolution



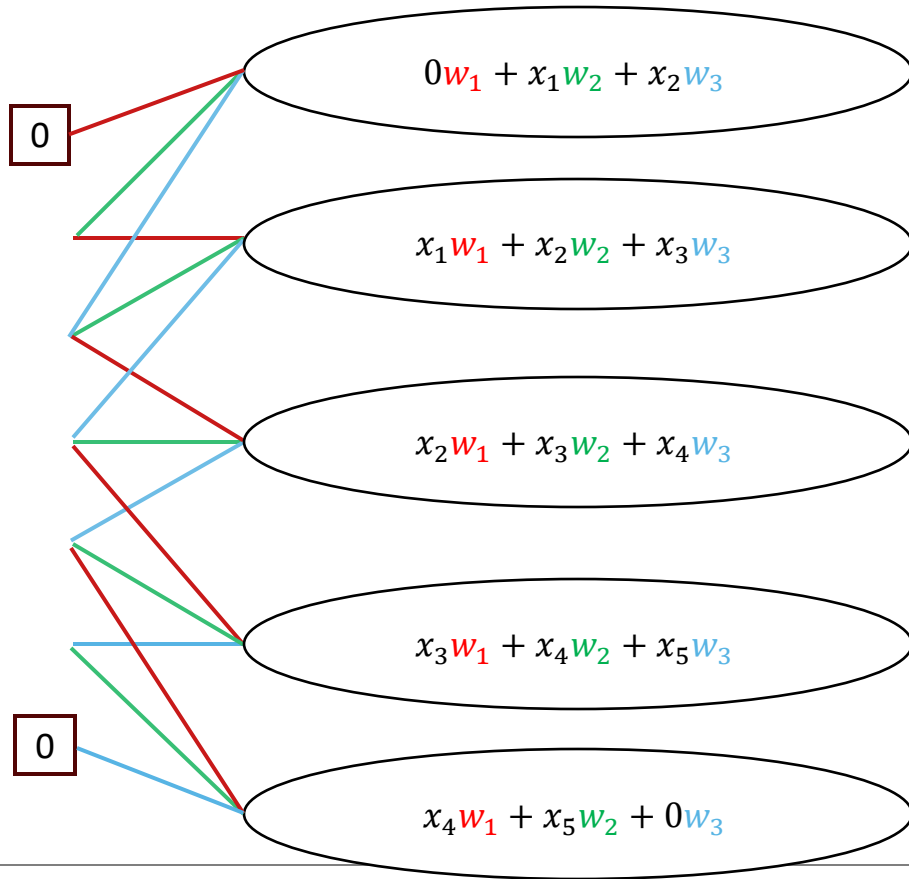


1-D Convolution

How can we keep the same number of features in hidden layer 1?

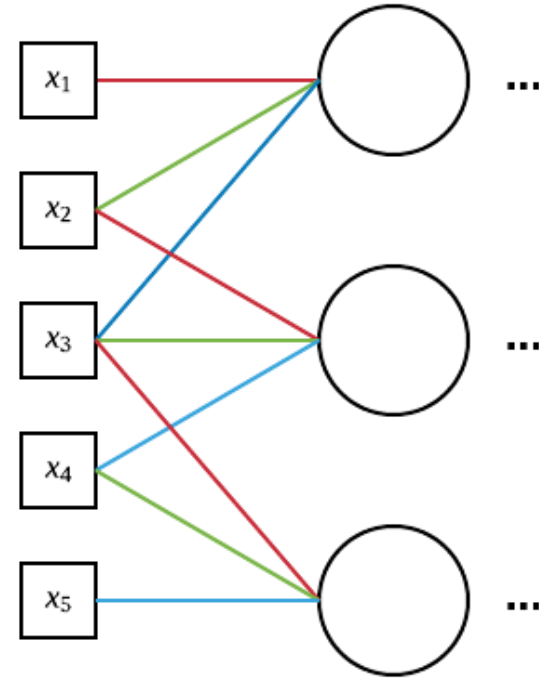
1-D Convolution

Zero-padding!



Properties of convolutional neural networks

- Sparse connectivity
- Weight sharing
- Parallel computations

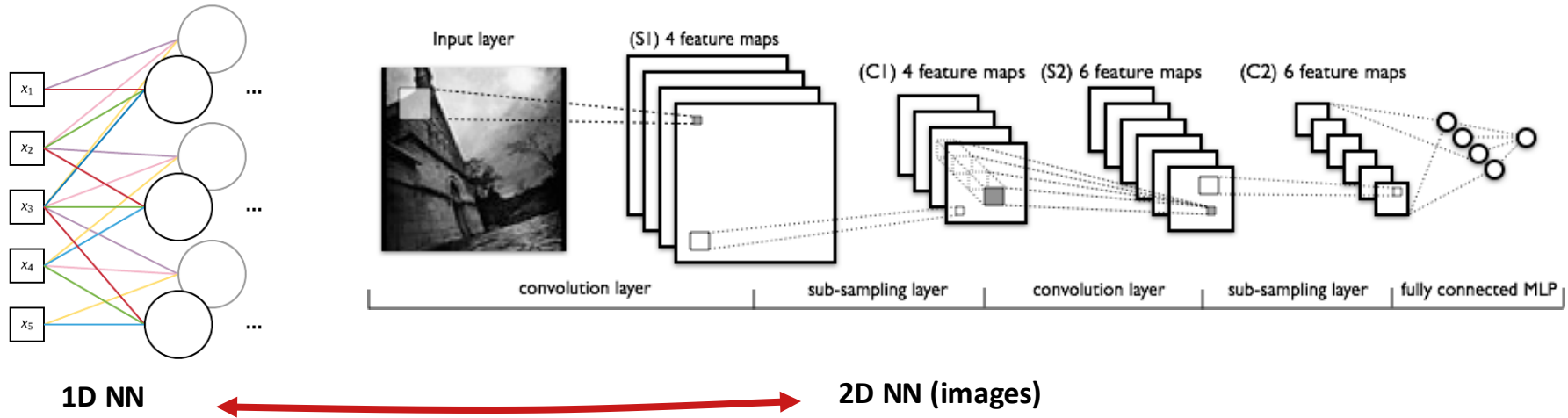


Why does this work?

- Local connectivity is effective for structured data (like images)
- The use of successive convolutional layers extract increasingly complex features. (Early layers detect simple features like edges and curves, while deeper layers learn more abstract concepts like shapes and objects.)

One added benefit is that the learned transformations will be equivariant with translation (if the features/image is shifted up/down the features will still be detected).

Convolutional neural networks are ideal for images



Because of the weight sharing, convolutional neural networks only make sense with structured data (such as images) as inputs.

I_1	I_2	I_3	I_4	I_5
I_6	I_7	I_8	I_9	I_{10}
I_{11}	I_{12}	I_{13}	I_{14}	I_{15}
I_{16}	I_{17}	I_{18}	I_{19}	I_{20}
I_{21}	I_{22}	I_{23}	I_{24}	I_{25}

*

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

=

$I_1 w_1 + I_2 w_2 + I_3 w_3$ $+$ $I_6 w_4 + I_7 w_5 + I_8 w_6$ $+$ $I_{11} w_7 + I_{12} w_8 + I_{13} w_9$		

I_1	I_2	I_3	I_4	I_5
I_6	I_7	I_8	I_9	I_{10}
I_{11}	I_{12}	I_{13}	I_{14}	I_{15}
I_{16}	I_{17}	I_{18}	I_{19}	I_{20}
I_{21}	I_{22}	I_{23}	I_{24}	I_{25}

*

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

=

$I_1 w_1 + I_2 w_2 + I_3 w_3$ $+$ $I_6 w_4 + I_7 w_5 + I_8 w_6$ $+$ $I_{11} w_7 + I_{12} w_8 + I_{13} w_9$	$I_2 w_1 + I_3 w_2 + I_4 w_3$ $+$ $I_7 w_4 + I_8 w_5 + I_9 w_6$ $+$ $I_{12} w_7 + I_{13} w_8 + I_{14} w_9$	

$$\begin{bmatrix} I_1 & I_2 & I_3 & I_4 & I_5 \\ I_6 & I_7 & I_8 & I_9 & I_{10} \\ I_{11} & I_{12} & I_{13} & I_{14} & I_{15} \\ I_{16} & I_{17} & I_{18} & I_{19} & I_{20} \\ I_{21} & I_{22} & I_{23} & I_{24} & I_{25} \end{bmatrix} * \begin{bmatrix} w_1 & w_2 & w_3 \\ w_4 & w_5 & w_6 \\ w_7 & w_8 & w_9 \end{bmatrix} =$$

$$\begin{bmatrix} I_1 w_1 + I_2 w_2 + I_3 w_3 + I_6 w_4 + I_7 w_5 + I_8 w_6 + I_{11} w_7 + I_{12} w_8 + I_{13} w_9 \\ I_2 w_1 + I_3 w_2 + I_4 w_3 + I_7 w_4 + I_8 w_5 + I_9 w_6 + I_{12} w_7 + I_{13} w_8 + I_{14} w_9 \\ I_3 w_1 + I_4 w_2 + I_5 w_3 + I_8 w_4 + I_9 w_5 + I_{10} w_6 + I_{13} w_7 + I_{14} w_8 + I_{15} w_9 \\ \hline \\ \hline \\ \hline \\ \hline \end{bmatrix}$$

$$\begin{array}{|c|c|c|c|c|} \hline I_1 & I_2 & I_3 & I_4 & I_5 \\ \hline I_6 & I_7 & I_8 & I_9 & I_{10} \\ \hline I_{11} & I_{12} & I_{13} & I_{14} & I_{15} \\ \hline I_{16} & I_{17} & I_{18} & I_{19} & I_{20} \\ \hline I_{21} & I_{22} & I_{23} & I_{24} & I_{25} \\ \hline \end{array} * \begin{array}{|c|c|c|} \hline w_1 & w_2 & w_3 \\ \hline w_4 & w_5 & w_6 \\ \hline w_7 & w_8 & w_9 \\ \hline \end{array} =$$

$$\begin{array}{|c|c|c|} \hline I_1 w_1 + I_2 w_2 + I_3 w_3 + I_6 w_4 + I_7 w_5 + I_8 w_6 + I_{11} w_7 + I_{12} w_8 + I_{13} w_9 \\ \hline I_2 w_1 + I_3 w_2 + I_4 w_3 + I_7 w_4 + I_8 w_5 + I_9 w_6 + I_{12} w_7 + I_{13} w_8 + I_{14} w_9 \\ \hline I_3 w_1 + I_4 w_2 + I_5 w_3 + I_8 w_4 + I_9 w_5 + I_{10} w_6 + I_{13} w_7 + I_{14} w_8 + I_{15} w_9 \\ \hline I_6 w_1 + I_7 w_2 + I_8 w_3 + I_{11} w_4 + I_{12} w_5 + I_{13} w_6 + I_{16} w_7 + I_{17} w_8 + I_{18} w_9 \\ \hline \\ \hline \\ \hline \\ \hline \\ \hline \end{array}$$

$$\begin{bmatrix} I_1 & I_2 & I_3 & I_4 & I_5 \\ I_6 & I_7 & I_8 & I_9 & I_{10} \\ I_{11} & I_{12} & I_{13} & I_{14} & I_{15} \\ I_{16} & I_{17} & I_{18} & I_{19} & I_{20} \\ I_{21} & I_{22} & I_{23} & I_{24} & I_{25} \end{bmatrix} * \begin{bmatrix} w_1 & w_2 & w_3 \\ w_4 & w_5 & w_6 \\ w_7 & w_8 & w_9 \end{bmatrix} =$$

$$\begin{bmatrix} I_1 w_1 + I_2 w_2 + I_3 w_3 + I_6 w_4 + I_7 w_5 + I_8 w_6 + I_{11} w_7 + I_{12} w_8 + I_{13} w_9 & I_2 w_1 + I_3 w_2 + I_4 w_3 + I_7 w_4 + I_8 w_5 + I_9 w_6 + I_{12} w_7 + I_{13} w_8 + I_{14} w_9 & I_3 w_1 + I_4 w_2 + I_5 w_3 + I_8 w_4 + I_9 w_5 + I_{10} w_6 + I_{13} w_7 + I_{14} w_8 + I_{15} w_9 \\ I_6 w_1 + I_7 w_2 + I_8 w_3 + I_{11} w_4 + I_{12} w_5 + I_{13} w_6 + I_{16} w_7 + I_{17} w_8 + I_{18} w_9 & I_7 w_1 + I_8 w_2 + I_9 w_3 + I_{12} w_4 + I_{13} w_5 + I_{14} w_6 + I_{17} w_7 + I_{18} w_8 + I_{19} w_9 & \\ & & \end{bmatrix}$$

$$\begin{array}{|c|c|c|c|c|} \hline I_1 & I_2 & I_3 & I_4 & I_5 \\ \hline I_6 & I_7 & I_8 & I_9 & I_{10} \\ \hline I_{11} & I_{12} & I_{13} & I_{14} & I_{15} \\ \hline I_{16} & I_{17} & I_{18} & I_{19} & I_{20} \\ \hline I_{21} & I_{22} & I_{23} & I_{24} & I_{25} \\ \hline \end{array} * \begin{array}{|c|c|c|} \hline w_1 & w_2 & w_3 \\ \hline w_4 & w_5 & w_6 \\ \hline w_7 & w_8 & w_9 \\ \hline \end{array} =$$

$$\begin{array}{|c|c|c|} \hline I_1 w_1 + I_2 w_2 + I_3 w_3 + I_6 w_4 + I_7 w_5 + I_8 w_6 + I_{11} w_7 + I_{12} w_8 + I_{13} w_9 \\ \hline I_2 w_1 + I_3 w_2 + I_4 w_3 + I_7 w_4 + I_8 w_5 + I_9 w_6 + I_{12} w_7 + I_{13} w_8 + I_{14} w_9 \\ \hline I_3 w_1 + I_4 w_2 + I_5 w_3 + I_8 w_4 + I_9 w_5 + I_{10} w_6 + I_{13} w_7 + I_{14} w_8 + I_{15} w_9 \\ \hline I_6 w_1 + I_7 w_2 + I_8 w_3 + I_{11} w_4 + I_{12} w_5 + I_{13} w_6 + I_{16} w_7 + I_{17} w_8 + I_{18} w_9 \\ \hline I_7 w_1 + I_8 w_2 + I_9 w_3 + I_{12} w_4 + I_{13} w_5 + I_{14} w_6 + I_{17} w_7 + I_{18} w_8 + I_{19} w_9 \\ \hline I_8 w_1 + I_9 w_2 + I_{10} w_3 + I_{13} w_4 + I_{14} w_5 + I_{15} w_6 + I_{18} w_7 + I_{19} w_8 + I_{20} w_9 \\ \hline \end{array}$$

$$\begin{array}{ccccc}
 I_1 & I_2 & I_3 & I_4 & I_5 \\
 I_6 & I_7 & I_8 & I_9 & I_{10} \\
 I_{11} & I_{12} & I_{13} & I_{14} & I_{15} \\
 I_{16} & I_{17} & I_{18} & I_{19} & I_{20} \\
 I_{21} & I_{22} & I_{23} & I_{24} & I_{25}
 \end{array}
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 w_4 & w_5 & w_6 \\
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 \begin{array}{c} I_2 w_1 + I_3 w_2 + I_4 w_3 \\ + \\ I_7 w_4 + I_8 w_5 + I_9 w_6 \\ + \\ I_{12} w_7 + I_{13} w_8 + I_{14} w_9 \end{array} &
 \begin{array}{c} I_3 w_1 + I_4 w_2 + I_5 w_3 \\ + \\ I_8 w_4 + I_9 w_5 + I_{10} w_6 \\ + \\ I_{13} w_7 + I_{14} w_8 + I_{15} w_9 \end{array} \\
 \begin{array}{c} I_6 w_1 + I_7 w_2 + I_8 w_3 \\ + \\ I_{11} w_4 + I_{12} w_5 + I_{13} w_6 \\ + \\ I_{16} w_7 + I_{17} w_8 + I_{18} w_9 \end{array} &
 \begin{array}{c} I_7 w_1 + I_8 w_2 + I_9 w_3 \\ + \\ I_{12} w_4 + I_{13} w_5 + I_{14} w_6 \\ + \\ I_{17} w_7 + I_{18} w_8 + I_{19} w_9 \end{array} &
 \begin{array}{c} I_8 w_1 + I_9 w_2 + I_{10} w_3 \\ + \\ I_{13} w_4 + I_{14} w_5 + I_{15} w_6 \\ + \\ I_{18} w_7 + I_{19} w_8 + I_{20} w_9 \end{array} \\
 \begin{array}{c} I_{11} w_1 + I_{12} w_2 + I_{13} w_3 \\ + \\ I_{16} w_4 + I_{17} w_5 + I_{18} w_6 \\ + \\ I_{21} w_7 + I_{22} w_8 + I_{23} w_9 \end{array} &
 \begin{array}{c} \\ \\ \\ \end{array} &
 \begin{array}{c} \\ \\ \\ \end{array}
 \end{array}$$

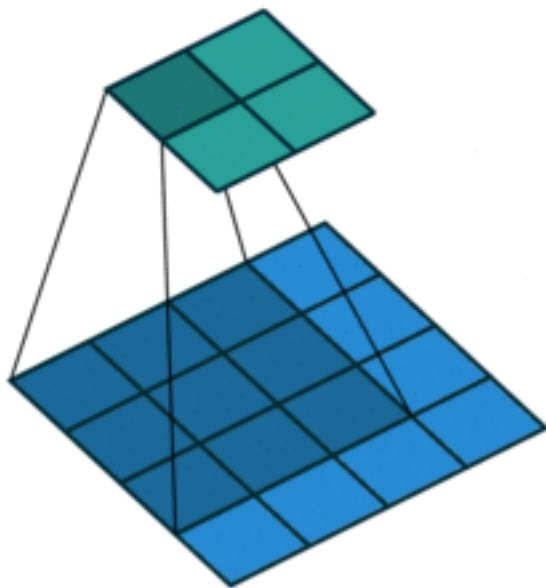
$$\begin{array}{ccccc}
 I_1 & I_2 & I_3 & I_4 & I_5 \\
 I_6 & I_7 & I_8 & I_9 & I_{10} \\
 I_{11} & I_{12} & I_{13} & I_{14} & I_{15} \\
 I_{16} & I_{17} & I_{18} & I_{19} & I_{20} \\
 I_{21} & I_{22} & I_{23} & I_{24} & I_{25}
 \end{array}
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 \begin{array}{ccc}
 w_1 & w_2 & w_3 \\
 w_4 & w_5 & w_6 \\
 w_7 & w_8 & w_9
 \end{array}
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$$\begin{array}{ccc}
 \begin{array}{c} I_1 w_1 + I_2 w_2 + I_3 w_3 \\ + \\ I_6 w_4 + I_7 w_5 + I_8 w_6 \\ + \\ I_{11} w_7 + I_{12} w_8 + I_{13} w_9 \end{array} &
 \begin{array}{c} I_2 w_1 + I_3 w_2 + I_4 w_3 \\ + \\ I_7 w_4 + I_8 w_5 + I_9 w_6 \\ + \\ I_{12} w_7 + I_{13} w_8 + I_{14} w_9 \end{array} &
 \begin{array}{c} I_3 w_1 + I_4 w_2 + I_5 w_3 \\ + \\ I_8 w_4 + I_9 w_5 + I_{10} w_6 \\ + \\ I_{13} w_7 + I_{14} w_8 + I_{15} w_9 \end{array} \\
 \begin{array}{c} I_6 w_1 + I_7 w_2 + I_8 w_3 \\ + \\ I_{11} w_4 + I_{12} w_5 + I_{13} w_6 \\ + \\ I_{16} w_7 + I_{17} w_8 + I_{18} w_9 \end{array} &
 \begin{array}{c} I_7 w_1 + I_8 w_2 + I_9 w_3 \\ + \\ I_{12} w_4 + I_{13} w_5 + I_{14} w_6 \\ + \\ I_{17} w_7 + I_{18} w_8 + I_{19} w_9 \end{array} &
 \begin{array}{c} I_8 w_1 + I_9 w_2 + I_{10} w_3 \\ + \\ I_{13} w_4 + I_{14} w_5 + I_{15} w_6 \\ + \\ I_{18} w_7 + I_{19} w_8 + I_{20} w_9 \end{array} \\
 \begin{array}{c} I_{11} w_1 + I_{12} w_2 + I_{13} w_3 \\ + \\ I_{16} w_4 + I_{17} w_5 + I_{18} w_6 \\ + \\ I_{21} w_7 + I_{22} w_8 + I_{23} w_9 \end{array} &
 \begin{array}{c} I_{12} w_1 + I_{13} w_2 + I_{14} w_3 \\ + \\ I_{17} w_4 + I_{18} w_5 + I_{19} w_6 \\ + \\ I_{22} w_7 + I_{23} w_8 + I_{24} w_9 \end{array} &
 \begin{array}{c} \\ \\ \\ \end{array}
 \end{array}$$

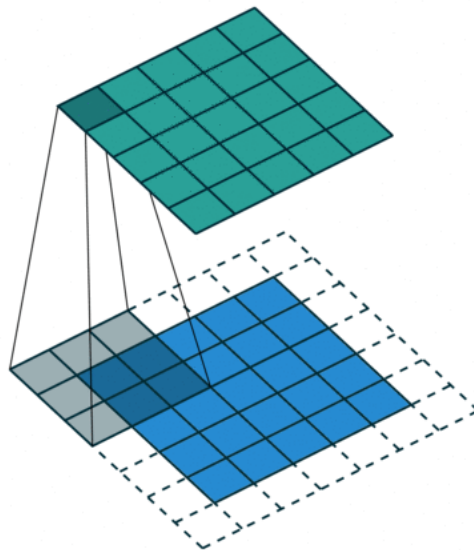
$$\begin{array}{ccccc}
 I_1 & I_2 & I_3 & I_4 & I_5 \\
 I_6 & I_7 & I_8 & I_9 & I_{10} \\
 I_{11} & I_{12} & I_{13} & I_{14} & I_{15} \\
 I_{16} & I_{17} & I_{18} & I_{19} & I_{20} \\
 I_{21} & I_{22} & I_{23} & I_{24} & I_{25}
 \end{array}
 *
 \begin{array}{ccc}
 w_1 & w_2 & w_3 \\
 w_4 & w_5 & w_6 \\
 w_7 & w_8 & w_9
 \end{array}
 + b =$$

$$\begin{array}{ccc}
 \begin{array}{c} I_1 w_1 + I_2 w_2 + I_3 w_3 \\ + \\ I_6 w_4 + I_7 w_5 + I_8 w_6 \\ + \\ I_{11} w_7 + I_{12} w_8 + I_{13} w_9 \end{array} &
 \begin{array}{c} I_2 w_1 + I_3 w_2 + I_4 w_3 \\ + \\ I_7 w_4 + I_8 w_5 + I_9 w_6 \\ + \\ I_{12} w_7 + I_{13} w_8 + I_{14} w_9 \end{array} &
 \begin{array}{c} I_3 w_1 + I_4 w_2 + I_5 w_3 \\ + \\ I_8 w_4 + I_9 w_5 + I_{10} w_6 \\ + \\ I_{13} w_7 + I_{14} w_8 + I_{15} w_9 \end{array} \\
 \begin{array}{c} I_6 w_1 + I_7 w_2 + I_8 w_3 \\ + \\ I_{11} w_4 + I_{12} w_5 + I_{13} w_6 \\ + \\ I_{16} w_7 + I_{17} w_8 + I_{18} w_9 \end{array} &
 \begin{array}{c} I_7 w_1 + I_8 w_2 + I_9 w_3 \\ + \\ I_{12} w_4 + I_{13} w_5 + I_{14} w_6 \\ + \\ I_{17} w_7 + I_{18} w_8 + I_{19} w_9 \end{array} &
 \begin{array}{c} I_8 w_1 + I_9 w_2 + I_{10} w_3 \\ + \\ I_{13} w_4 + I_{14} w_5 + I_{15} w_6 \\ + \\ I_{18} w_7 + I_{19} w_8 + I_{20} w_9 \end{array} \\
 \begin{array}{c} I_{11} w_1 + I_{12} w_2 + I_{13} w_3 \\ + \\ I_{16} w_4 + I_{17} w_5 + I_{18} w_6 \\ + \\ I_{21} w_7 + I_{22} w_8 + I_{23} w_9 \end{array} &
 \begin{array}{c} I_{12} w_1 + I_{13} w_2 + I_{14} w_3 \\ + \\ I_{17} w_4 + I_{18} w_5 + I_{19} w_6 \\ + \\ I_{22} w_7 + I_{23} w_8 + I_{24} w_9 \end{array} &
 \begin{array}{c} I_{13} w_1 + I_{14} w_2 + I_{15} w_3 \\ + \\ I_{18} w_4 + I_{19} w_5 + I_{20} w_6 \\ + \\ I_{23} w_7 + I_{24} w_8 + I_{25} w_9 \end{array}
 \end{array}
 + b$$

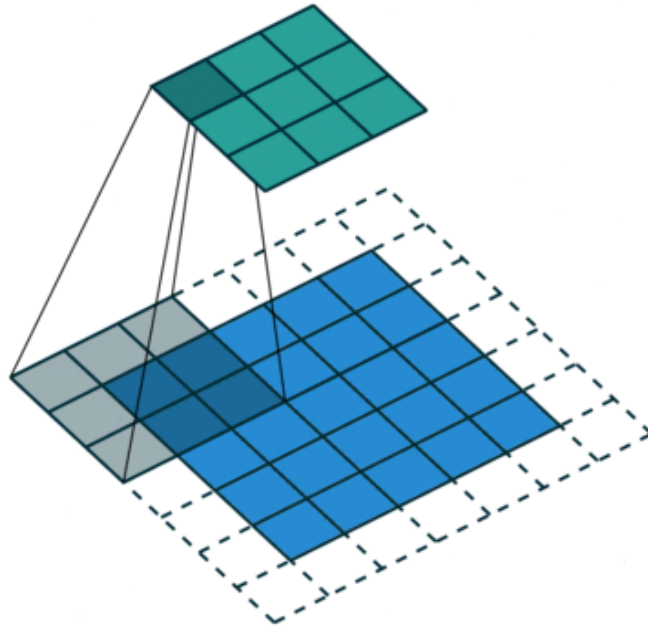
2D convolution



Padding



Stride

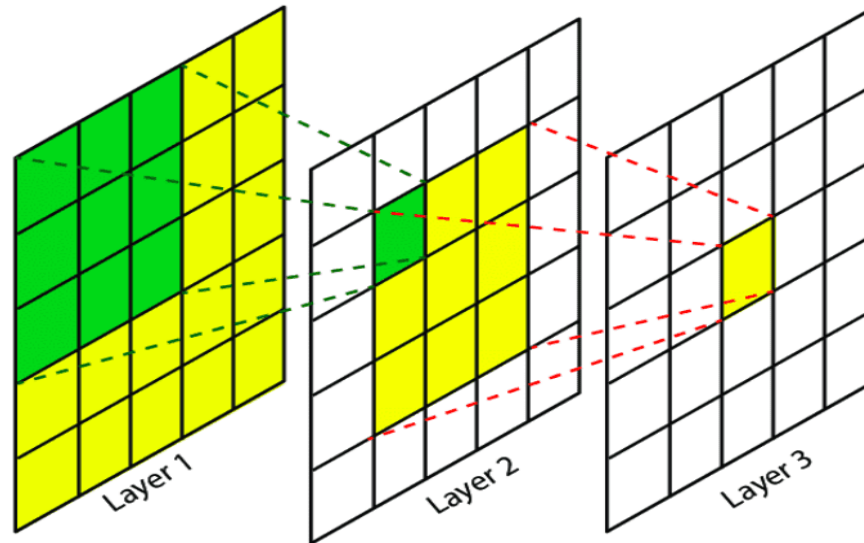


In general

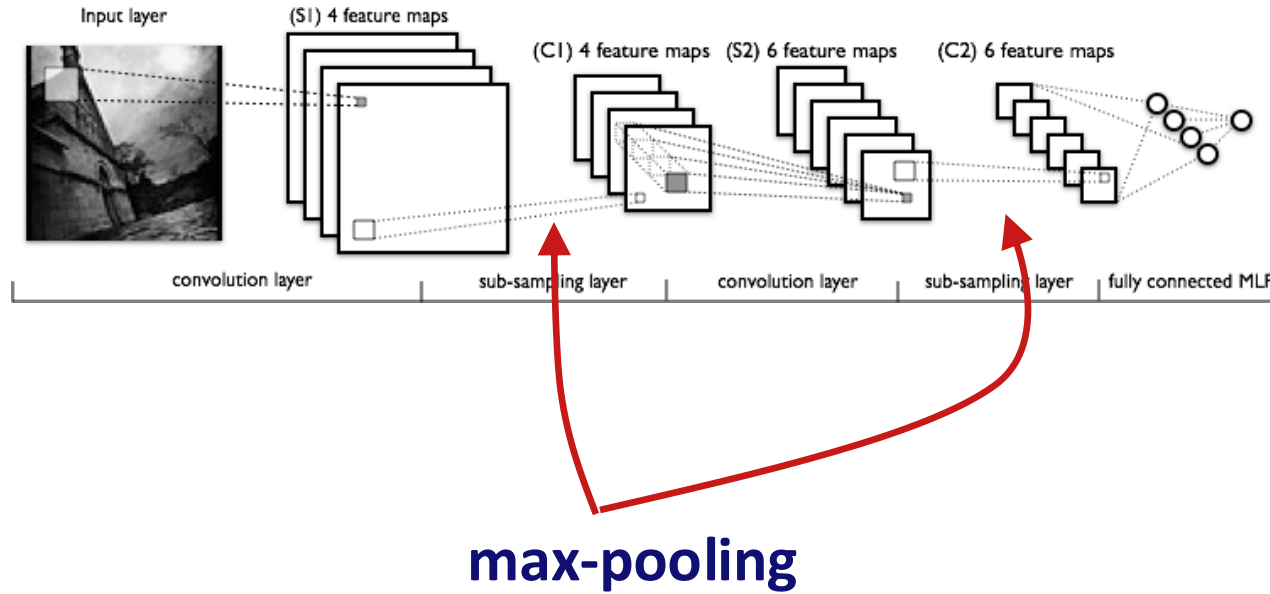
$$\text{Output_size} = (((\text{input_size} - \text{kernel_size} + 2 * \text{padding}) / \text{stride}) + 1)$$

Receptive field

Receptive Field is defined as the size of the region in the input that produces the feature

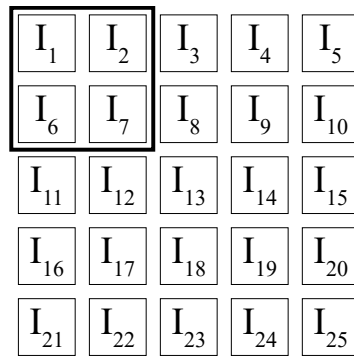


Max-pooling

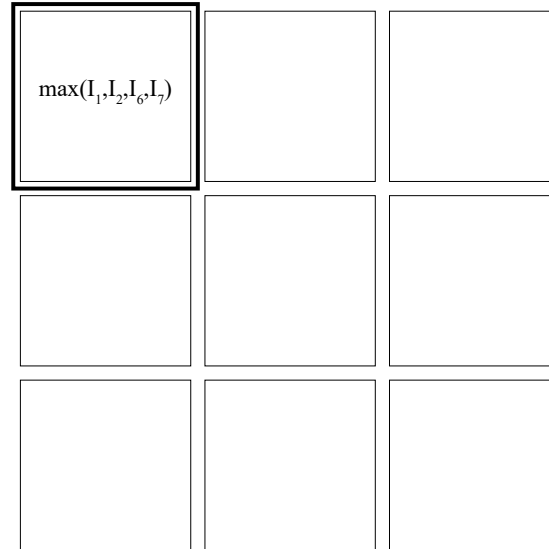


Max-pooling

- Reduce size of feature space
- Maximum of features
- Typical kernel size = 2×2



2×2 max pooling

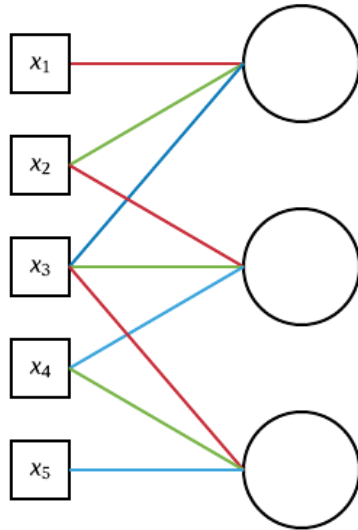


Benefits of max-pooling

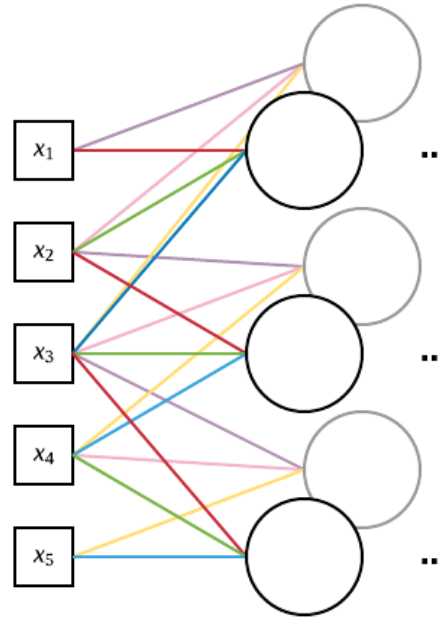
- “Quickly” reduces the number size of the feature maps
- Introduces translation invariance (slightly translated version of the input image will result in the same output)

Alternative: Average Pooling

Depth



shared weights
3 weights

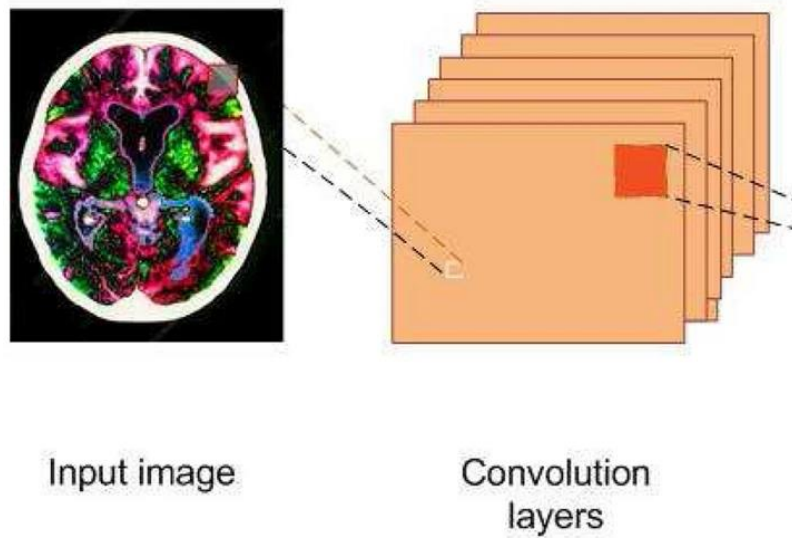


two sets shared weights
6 weights

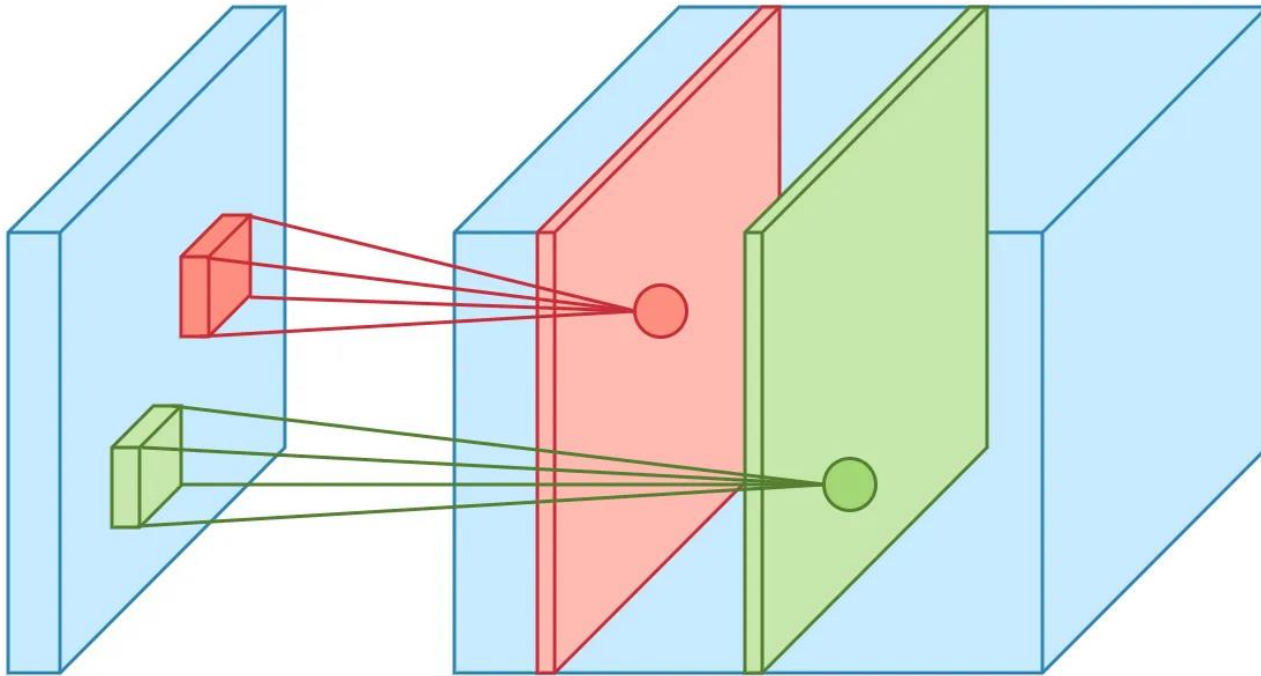
We can add additional sets of weights that can learn additional interesting transformation of the input.

Note that the added neurons are not a new layer. They are part of layer 1.

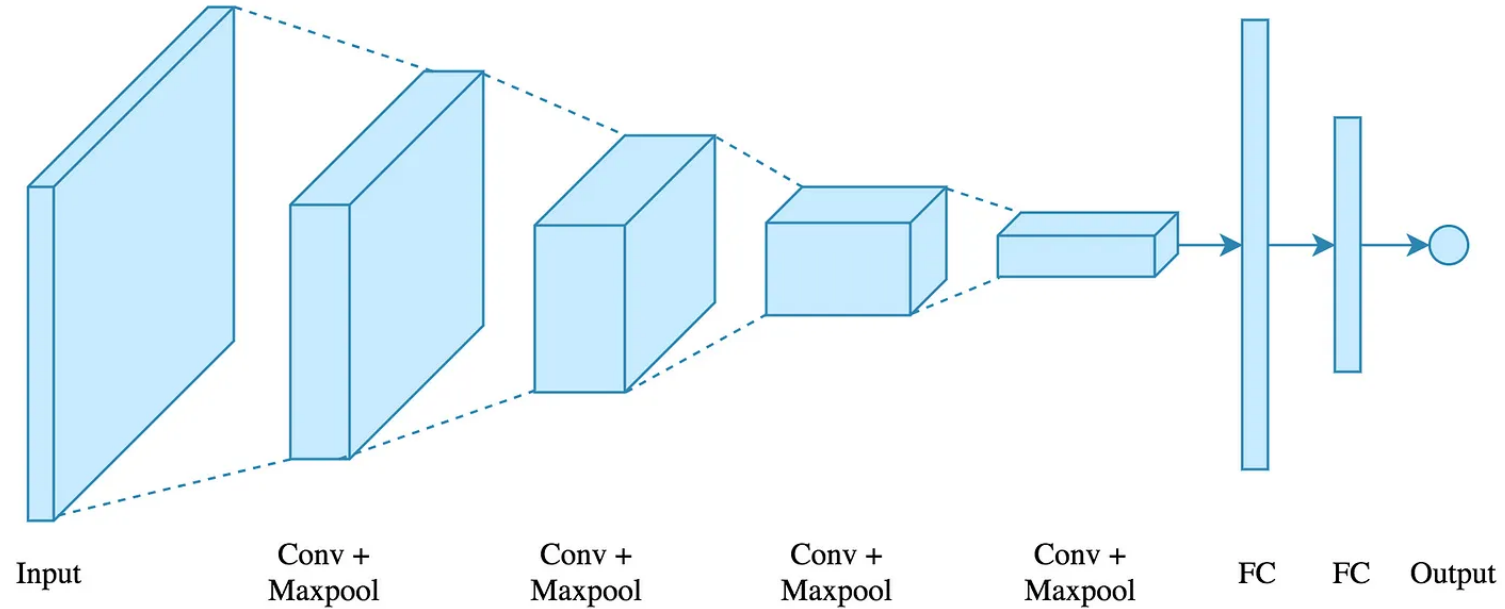
Depth



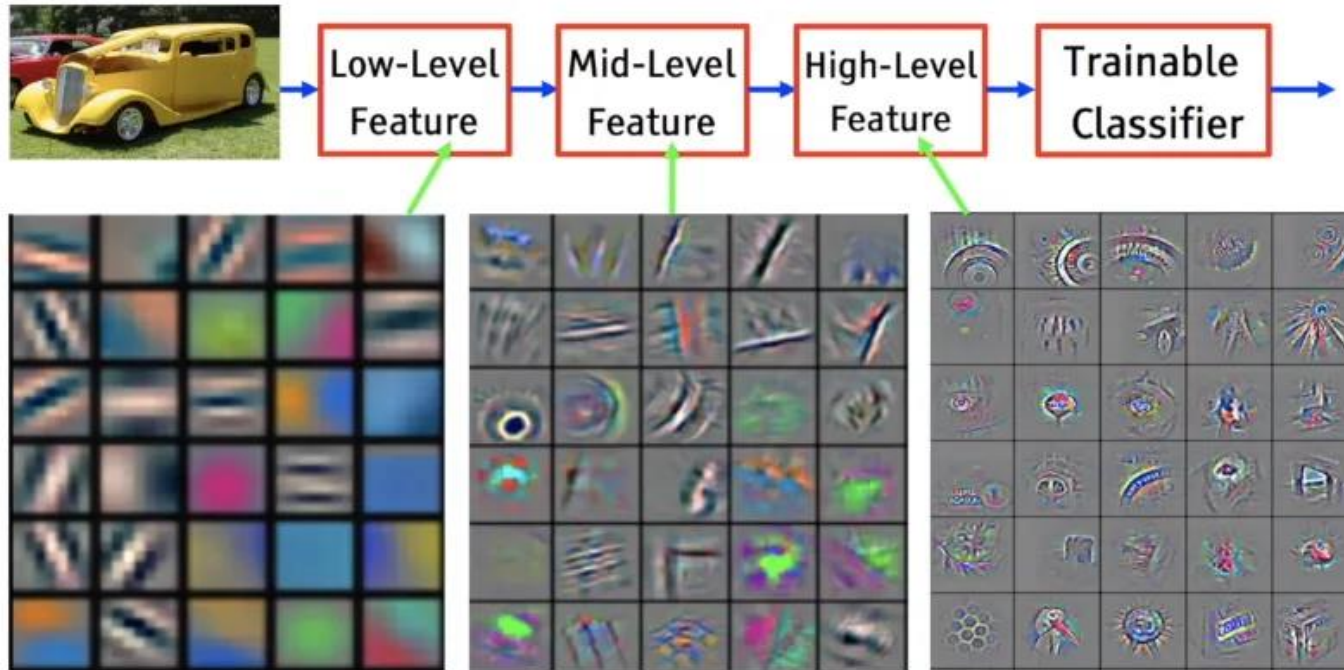
Channel depth



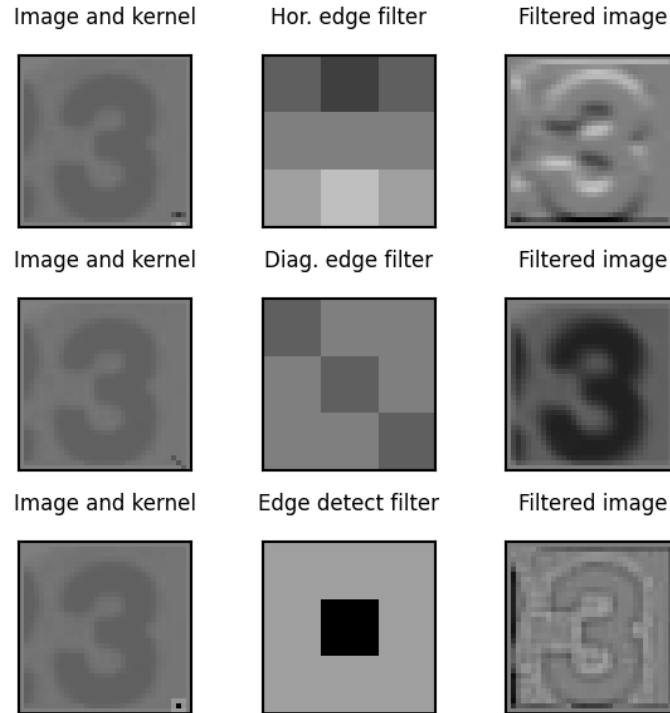
CNN



Convolutional feature extractors



Convolutional feature extractors (relation to 8BB050)



Summary

- Concept of **convolutions** in a neural network
- Why can we use a **convolutional approach** for (medical) **images**
- Convolutions enable development of **deep** (and large) **neural networks**
- **Max-pooling layer** in a convolutional neural network
- **Kernel size**
- **Receptive field**

Questions?